

A Minor Project Inception Report on

VoteChain

Submitted in Partial Fulfillment of the Requirements for

The Degree of Bachelor of Engineering in Computer Engineering

Under Pokhara University

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Department of Computer Engineering
UNIVERSAL ENGINEERING AND SCIENCE COLLEGE

INCEPTION REPORT

Project Title:

VoteChain: Secure Blockchain Voting System

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Team Members:

Abhash Basnet – Frontend Development
Sandesh Paudel – Backend Development

Achieve Paudel – Blockchain Development

Rajib Raut- Blockchain Development
Ayush Jasraj – Testing and Quality Assurance

1. Introduction

The reliability of electoral processes is fundamental to democracy. However, conventional voting systems are often prone to issues such as vote tampering, low transparency, delayed results, and lack of trust. VoteChain aims to address these issues by leveraging blockchain technology to create a tamper-proof, transparent, and efficient voting system. It integrates biometric verification, smart contracts, and real-time vote tallying to provide a secure and user-friendly digital voting experience.

2. Objectives

The primary objectives of VoteChain are:

- To design a secure electronic voting system using blockchain and smart contracts.
- To implement biometric (fingerprint) voter authentication and Election ID login.

- To enable real-time vote casting and tallying with transparency.
- To ensure immutability and auditability of the vote ledger.
- To develop an admin dashboard for managing elections and overseeing system activity.

3. Scope of the Project

- Core System: Blockchain-based backend infrastructure to store votes and handle smart contracts.
- User Modules:
 - Voter Interface: For secure login, fingerprint authentication, and vote casting.
 - Admin Interface: For election creation, voter registration, and result monitoring.
- Security Measures:
 - Fingerprint validation using external biometric modules.
 - Hashing of all sensitive data.
 - Role-based access controls for different users.
- Blockchain: Use of private Ethereum-based blockchain (e.g., Ganache, Polygon, or testnet) to store immutable vote records.

4. Methodology

Phase 1: Research and Design – Study of blockchain frameworks, system design, UI/UX wireframes, requirement gathering

Phase 2: Development – Module-wise development (frontend, backend, biometric integration, blockchain smart contracts)

Phase 3: Testing – Functional testing, vulnerability testing, test cases on vote casting and validation

Phase 4: Deployment – Deployment on private blockchain test network with mock data

Phase 5: Report & Documentation – Final report writing, user manual, presentation slides

5. Tools and Technologies

Frontend: ReactJS / HTML-CSS-JS with Tailwind

Backend: Node.js / Express.js

Blockchain: Custom with POA/POS/PBFT

Biometric: DigitalPersona / R305 fingerprint module via Arduino or SDK

Database: MongoDB / IPFS for some parts

Other Tools: Git, Figma (UI design), Postman (API testing), Metamask (Wallet Integration)

6. Work Division

Achieve : Project design, smart contract integration, biometric module, reporting, core development.

Rajib: Core Development

Abhash: Frontend development (UI, forms, voting interface)

Sandesh: Backend integration, blockchain logic, server setup

Ayush: Test case design, system testing, documentation support

Timeline

Task	Date
Research & Planning	Sept 17 – Oct 5, 2025
Development Phase 1 (Frontend/Backend)	Oct 6 – Nov 15, 2025
Blockchain Integration & Testing	Nov 16 – Dec 10, 2025
Final Testing, Optimization	Dec 11 – Jan 10, 2026
Documentation & Submission	Jan 11 – Feb 5, 2026

(Timeline may be adjusted around the exam schedule)

8. Expected Outcomes

- A working prototype of a Blockchain-based secure voting platform.
- A fully documented system with:
 - Real-time vote tallying
 - Immutable ledger of votes
 - Fingerprint authentication
 - Admin dashboard
- Academic research contribution to blockchain use in governance.
- Potential for real-world pilot testing in campus or small-scale elections.

9. Limitations

- Limited real-world biometric device access for large-scale testing.
- Network latency in blockchain simulations on local testnets.
- Usability constraints on fingerprint hardware compatibility with web apps.

10. Conclusion

VoteChain is a timely and impactful project aiming to demonstrate how blockchain and biometrics can revolutionize electoral systems. With proper execution, the system can serve as a prototype for national or organizational e-voting platforms, ensuring trust, transparency, and security.